

Igloo: Making insurance simple and affordable via big data and machine learning on Google Cloud

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[About Igloo](#)



Igloo provides more affordable insurance, faster approvals, models for pricing management, processes millions of insurance on BigQuery to speed underwriting and

About Igloo

Founded in 2016, Igloo (formerly known as Axinan) is a licensed InsurTech firm that aims to offer simple and affordable insurance to everyone. It offers a wide portfolio of insurance products ranging from personal to lifestyle insurance. Leveraging big data and machine learning, Igloo creates on-demand and usage based products for a digital ecosystem that includes ecommerce platforms, mobile operators, banks, and travel websites. Operating in Australia, Indonesia, Philippines, Malaysia,

Google Cloud results

- Automates 80% to 90% of low-risk quotes and claims so insurance officers can invest time in high-risk cases
- Processes 5 million policies each month to protect customer lifestyle in a digital economy
- Processes millions of insurance data with zero downtime to keep partners

Custom quotes automated under 5 minutes

Singapore, Thailand, and Vietnam, Igloo has sold 80 million policies to date.

Industries: Financial Services & Insurance

Location: Singapore

Cloud Bigtable

Google Cloud (<https://cloud.google.com/>) Cloud SQL (<https://cloud.google.com/sql/>), Looker

BigQuery (<https://cloud.google.com/bigquery/>) (<https://marketingplatform.google.com/about/d>

Google Workspace (<https://workspace.google.com/>)

Kubernetes Engine

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Cloud SQL (<https://cloud.google.com/sql/>)

(<https://cloud.google.com/automl/>), Cloud

Looker Studio (<https://marketingplatform.google.com/about/data-studio/>)

(<https://workspace.google.com/products/drive/>)
Cloud IAM (<https://cloud.google.com/iam/>)

Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Cloud Spanner (<https://cloud.google.com/spanner/>)
Google Meet

Vision AI (<https://cloud.google.com/vision/>) Google Meet

Google Drive (<https://workspace.google.com/products/drive/>)

Docs (<https://workspace.google.com/products/docs/>)

and their
online
customers
happy

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Did you know that millennials are twice as likely

(<https://news.gallup.com/businessjournal/181829/insurance-companies-big-problem-millennials.aspx>)

as other generations to buy insurance policies online? To meet the needs of this growing market, Igloo (<https://www.iglooinsure.com>) uses big data and machine learning (ML) to create digital products that are more relevant to millennial lifestyles, such as gadget insurance.

Igloo also works with digital partners such as ecommerce platforms to engage consumers where they shop. Consumers can buy insurance directly on the check-out page, without filling out paperwork or speaking to an agent.

Working with other digital platforms gives Igloo the opportunity to collect and analyze vast amounts of non-traditional data such as consumer behavior data from ecommerce platforms for actionable insights. It also runs ML models on Google Cloud (<https://cloud.google.com/>) to turn buying a policy or filing a claim into a fuss-free process.

Igloo's competitive edge lies in big data and ML. To accelerate growth, it's replicating successful use cases from fintechs in more developed markets, with minor adjustments to fit insurance products and local customer demand for new partners in

different countries. Google Cloud helps it scale easily for a high-frequency model where some partners receive millions of page views each day.

Igloo's insurtech applications are available to third-party developers through APIs deployed in [Google Kubernetes Engine](#) (/kubernetes-engine). For data and analytics, Igloo uses [BigQuery](#) (/bigquery) for its data warehouse, [Cloud SQL](#) (/sql) for transactional data, [Cloud Spanner](#) (/spanner) for regional configurations, and [Cloud Bigtable](#) (/bigtable) for storing IoT data. For its AI stack, Igloo uses [AutoML Vision](#) (/vision/automl/docs) to automate claim processing.

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—Jack Xia, Chief Actuary,
Igloo

Transforming insurance with data insights

Igloo processes five million policies each month, from transit insurance for ecommerce platforms to gadget insurance for mobile operators. So how does it determine the price of each insurance policy?

When a customer requests a quote on the partner’s checkout page, Igloo loads dozens of data points into the system for risk analysis with AI. In the case of transit insurance covering goods against loss and damage during delivery, Igloo may look at risk factors such as pricing, product description, and product photos. For electronic gadget insurance, Igloo may consider user demographics, recent claims, and information about the product, such as brand and model.

For now, Igloo's health insurance is pretty straightforward. If the applicant has no claim history, insurers set premiums based on age and gender.

On the other hand, consumer data from ecommerce platforms is far more complex. Apart from price, Igloo also processes multidimensional data such as customer lifetime value, product name, product categories, product descriptions, images, and so on.

"We run sophisticated ML models to find complex patterns from the data and make accurate risk assessments to calculate the right premium for each policy," says Jack Xia, Chief Actuary at Igloo. "Thanks to Google Cloud, our data analysts can utilize managed services and focus on developing new digital insurance products, instead of maintaining the data warehouse."

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Collaborating with external partners to expand market reach

Traditional insurance companies tend to support a monolithic architecture. This often slows down innovation because of legacy backend systems. In comparison, Igloo adopts a microservices architecture on Google Kubernetes Engine to quickly

respond to customer needs and market opportunities.

“By building containerized applications on Google Kubernetes Engine, we can easily integrate our technology with different partners,” says Jack.

“We’re helping traditional insurers activate their digital capabilities, without investing in underlying infrastructure. For ecommerce partners, it means providing an additional touchpoint to build customer loyalty.”

It reuses common components such as user profile management, authentication service, policy information, and claims processing for each partner deployment, so partners can easily integrate Igloo’s products on their platform through secure APIs, instead of coding sophisticated insurance systems from scratch.

Igloo also uses custom-built risk engines to calculate the premium based on each consumer’s preferences and behavior. The lower the risks, the lower the premiums. Jack explains that low-risk policies such as gadget insurance cost less than one cent a day. The company is now automating its low-risk quotes and claims so that insurance officers can focus on reviewing the high-risk cases instead. “For 80% to 90% of low-risk claims, customers get it approved in less than a second without requiring a human in the loop,” says Jack.

Because Igloo fully integrates with platforms, it needs to ensure high availability at all times, especially during mega sales events. “Our load

increases by five to 10 times when ecommerce partners hold mega sales events such as Hari Raya and Singles Day,” says Jack, adding that the platform needs to be able to run millions of database processes with zero downtime during peak periods. “Google Kubernetes Engine helps us scale resources so we focus on coding instead of managing servers.”

“Looker Studio integrates natively with BigQuery and other Google databases to display data from different sources in interactive dashboards. This enables Igloo to share actionable insights with our partners, optimise current products and

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—Raunak Mehta, Chief
Commercial Officer, Igloo

Simplifying data insights for partners

For reporting purposes, Igloo provides partners with a bulk data transfer weekly or monthly, depending on the partner requirement. Partners can also assess and download real-time data from Igloo's dashboard on Looker Studio (<https://marketingplatform.google.com/about/data-studio/>) at any time.

“Looker Studio integrates natively with BigQuery and other Google databases to display data from different sources in interactive dashboards. This enables Igloo to share actionable insights with our partners, optimize current products, and identify opportunities for new ones,” says Raunak Mehta, Chief Commercial Officer at Igloo.

Partners don't need in-depth technical knowledge to make sense of data in Looker Studio. Product managers can create reports using filters such as user demographics, date range, and audience type to drill down on data at a granular level. These

insights help partners tailor marketing campaigns or pricing strategies for different target audiences.

AI for no-fuss claims

Catering to its mobile-savvy customers, Igloo enables people to make claims directly through its app on their mobile devices with the help of AI. This saves customers' precious time waiting in line for a claim officer. For example, a consumer covered by Igloo's gadget insurance can submit an accidental damage claim for the product through their phone, simply by sending a picture of the damage as proof. Igloo uses Cloud Vision (/vision) to identify the product submitted by comparing it with the information of the product in the policy and on the ecommerce website.

Using AutoML Vision, Igloo compares photos of the damaged product with pictures at the start of the policy for an estimated repair cost. For significant damages, Igloo appoints an appraiser to follow up with a careful physical inspection.

"Thanks to AutoML Vision, we can quickly analyze damages and deliver a preliminary estimate of damages to consumers to speed up the claim process," says Jack. "This self-service process improves customer experience and allows appraisers to save time from initial assessments."

Easy integration with third-party applications

Internally, Igloo has been on Google Workspace (<https://workspace.google.com/>) (formerly G Suite) since the day it launched. Currently, more than 50 employees across Igloo's offices in Southeast Asia use productivity and collaboration apps such as Gmail (<https://workspace.google.com/products/gmail/>), Drive (<https://workspace.google.com/products/drive/>), Docs (<https://workspace.google.com/products/docs/>), Sheets (<https://workspace.google.com/products/sheets/>), and Google Meet (<https://workspace.google.com/products/meet/>).

"Instead of sending files back and forth over email, team members can edit the same document or spreadsheet on Google Workspace at the same time," says Jack. "This is especially valuable during COVID-19 with employees working remotely."

Igloo also uses Drive API (<https://developers.google.com/drive/api/v3/about-sdk>) to retrieve data from third-party sources and display it in Sheets for analysis. The data scientists then write and modify Python code in Sheets to speed up ML projects.

"Employees also save time by using Google login for third-party services such as Slack and Confluence," Jack adds. "We can assign granular access control with Cloud Identity and Access Management (/iam) so employees can access the right amount of data to get their work done."

Unlocking IoT data opportunities for future growth

Igloo is running a pilot project for car insurance as a new product offering for 2020. Looking beyond traditional indicators such as age and gender, the new product will analyze real-time vehicle data on Bigtable to assess the individual's driving behavior and patterns.

"We're trying to expand our product lines into smart homes, IoT, and cyber insurance. And all these solutions require real-time monitoring of the underlying assets, whether it's a camera to detect any abnormal movements in your home or GPS locator in a car for location speed," says Jack.

"Google Cloud gives us the computing power and processing capabilities to gather and analyze rich data to innovate and grow our business."